

Student Grade Portal - REST API Endpoints Guide

All API endpoints are hosted relative to the base URL: `http://localhost:8081` . Every API call except authentication endpoints requires a valid JWT token sent in the headers as: `Authorization: Bearer <JWT_TOKEN>` .

1. Authentication Endpoints

Exposed in AuthController.java.

Log In

- Method / Route: `POST /api/auth/login`
- Access level: Public
- Request Body:

```
{
  "username": "admin",
  "password": "admin123"
}
```

- Response (HTTP 200):

```
{
  "token": "eyJhbGciOiJIUzI1...",
  "username": "admin",
  "role": "ADMIN"
}
```

- Description: Validates credentials. Returns the JWT token to use in subsequent authorization headers.

Register System Account

- Method / Route: `POST /api/auth/register`
- Access level: `ADMIN` only
- Request Body:

```
{
  "username": "new_teacher",
  "password": "securepassword123",
  "role": "TEACHER"
}
```

- Description: Creates a new credentials profile to log in to the portal. The role parameter accepts `ADMIN` , `TEACHER` , or `STUDENT` .

Change My Password

- Method / Route: `POST /api/auth/change-password`
- Access level: Authenticated (Any Role)

- Request Body:

```
{
  "oldPassword": "oldpassword123",
  "newPassword": "newsecurepassword123"
}
```

- Description: Updates the password of the user currently authenticated by the header token.

2. Student Management Endpoints

Exposed in StudentController.java.

Get Student Directory

- Method / Route: `GET /api/students`
- Access level: Authenticated (Behavior changes by role)
 - *Admin / Teacher*: Returns a list of **all** student profiles in the database.
 - *Student*: Returns a list containing **only** the logged-in student's profile record.

Get Student by ID

- Method / Route: `GET /api/students/{id}`
- Access level: Authenticated (Behavior changes by role)
 - *Admin / Teacher*: Can view any student's record details.
 - *Student*: Can only view their own profile. Attempting to view another ID returns `403 Forbidden`.

Create Student Profile

- Method / Route: `POST /api/students`
- Access level: `ADMIN` or `TEACHER`
- Request Body:

```
{
  "firstName": "Jane",
  "lastName": "Smith",
  "email": "janesmith@pasadena.edu",
  "phoneNumber": "626-555-0210",
  "address": "100 N Garfield Ave, Pasadena, CA 91101"
}
```

- Description: Creates a new student record. The `email` field must be unique.

Update Student Profile

- Method / Route: `PUT /api/students/{id}`
- Access level: `ADMIN` or `TEACHER` (Conditional constraints)
 - *Admin*: Can modify all fields.
 - `403 Forbidden`
- Description: Updates details for a specific student.

Delete Student Profile

- Method / Route: `DELETE /api/students/{id}`
- Access level: `ADMIN` only
- Description: Removes a student profile from the system. Automatically deletes all linked student grade records and their credential login account.

3. Grade & Scorecard Endpoints

Exposed in GradeController.java.

Get Grade Records Directory

- Method / Route: `GET /api/grades`
- Access level: `ADMIN` or `TEACHER` (Students are blocked with `403 Forbidden`)
- Description: Returns a full list of all grade records in the system.

Get Grades for Specific Student

- Method / Route: `GET /api/grades/student/{studentId}`
- Access level: Authenticated (Behavior changes by role)
 - *Admin / Teacher*: Can view grading scorecards for any student ID.
 - *Student*: Can only query their own student ID. Accessing another student's ID returns `403 Forbidden` .

Modify Grade & Feedback

- Method / Route: `PUT /api/grades/{id}`
- Access level: `ADMIN` or `TEACHER`
- Request Body:

```
{
  "score": 92.5,
  "feedback": "Outstanding assignment review."
}
```

- Description: Updates a grade score. Score values must be non-negative and cannot exceed the assignment's maximum allowed points.

Bulk Import Grades (CSV)

- Method / Route: `POST /api/grades/import`
- Access level: `ADMIN` or `TEACHER`
- Request Form-Data: `file` (multipart CSV spreadsheet file containing grading metadata columns).
- Description: Automatically generates/links users, students, courses, assignments, and updates grade values. Returns a detailed execution log list.

4. User Directory Control Endpoints

Exposed in UserController.java.

Get Users Directory

- Method / Route: `GET /api/users`
- Access level: `ADMIN` only
- Description: Returns a list of all system credentials accounts.

Get User Details

- Method / Route: `GET /api/users/{id}`
- Access level: `ADMIN` only
- Description: Retrieves details for a specific user login profile.

Update User Details

- Method / Route: `PUT /api/users/{id}`
- Access level: `ADMIN` only
- Request Body:

```
{
  "password": "optional_new_password",
  "role": "TEACHER"
}
```

- Description: Modifies credentials password or role type.

Delete User Account

- Method / Route: `DELETE /api/users/{id}`
- Access level: `ADMIN` only
- Description: Wipes login access credentials from the system.